

Appendix to the final project report

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1 Part one

Parameter	Description	Unit
e_{ij}	Edge between nodes i and j	-
c_{ij}^{fix}	Fixed costs on e_{ij}	€/m
c_{ij}^{var}	Variable costs on e_{ij}	€/m.kW
c_{ij}^{om}	Maintenance costs on e_{ij}	€/year
c^{heat}	Heat generation cost at the source	€/kWh
c^{rev}	Revenue on the heat delivered	€/kWh
θ_{ij}^{fix}	Fixed heat losses on e_{ij}	kW/m
θ_{ij}^{var}	Variable heat losses on e_{ij}	kW/kW.m
l_{ij}	Length of pipe e_{ij}	m
d_i	Potential peak demand at node i	kW
D_i	Annual demand at node i	kWh/year
C_{ij}^{max}	Maximum capacity of pipe e_{ij}	kW
Q^{max}	Maximum capacity the source can deliver	kW
μ_t	Percentage of edges required for a lot of type t	-
N_{it}	Parameter equal to 1 if node i is a building of type t , 0 otherwise	-
α	Sales factor for the investment costs	-
λ	Connection quota	-
β	Competition effect	-

Table 1: Parameters of the mathematical model

Variables

- $V = \{1, \dots, n\}$: set of nodes.
- v_0 : index of the source node.
- X_{ij} : binary variable indicating whether a pipe e_{ij} is used or not
- p_{ij}^{in} power entering pipe e_{ij}
- p_j^{out} power leaving pipe e_{ij}

Building the objective function

Distances: $l_{ij} = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2} \quad \forall i, j \in V$

Demand: $D_j = S_j \times SC_j$ where: $\begin{cases} S_j : & \text{Surface of node } j \text{ (m}^2\text{)} \\ SC_j : & \text{Surface_consumer of node } j \text{ (kWh/m}^2\text{/a)} \end{cases}$

Revenue: $\sum_{i,j \in V} c^{rev} D_j \lambda X_{ij}$

Heat generation costs: $\sum_{\substack{j \in V \\ j \neq v_0}} X_{v_0 j} D_j c^{heat} \lambda$

Maintenance costs: $\sum_{i,j} X_{ij} c_{ij}^{om} l_{ij}$

Fixed costs: $\sum_{i,j} \alpha X_{ij} l_{ij} c_{ij}^{fix}$

Variable costs: $\sum_{i,j} \alpha c_{ij}^{var} l_{ij} X_{ij} p_{ij}^{in}$

Objective function

The objective function Z to minimize is therefore:

$$Z = \sum_{i,j} X_{ij} l_{ij} \left[c_{ij}^{om} + \alpha \left(c_{ij}^{fix} + c_{ij}^{var} p_{ij}^{in} \right) \right] + \sum_{j \in V} \lambda X_{0j} c^{heat} D_j - \sum_{j \in V} c^{rev} D_j \lambda X_{0j}$$

Constraints

Tree structure:
$$\begin{cases} \forall j \in V, & \sum_{\substack{i \neq v_0 \\ i \neq j}} X_{ji} \leq 1 \\ \forall i \in V, & X_{ii} = 0 \end{cases}$$

Uniqueness: $\forall i \neq j \in V, \quad X_{ij} + X_{ji} \leq 1$

Demand satisfaction:

- **Losses:** $\eta_{ij} = 1 - \theta_{ij}^{out} P_{ij}$

- **Usage loads:** $\delta_{ij} = d_j \beta \lambda + \theta_{ij}^{fix} l_{ij}$

- **Outgoing flow:** $p_{ij}^{out} = \eta_{ij} p_{ij}^{in} - \delta_{ij} X_{ij}$

Balance / Energy conservation: $\forall j \neq v_0, \quad \sum_{\substack{i \in V \\ i \neq j}} p_{ij}^{in} - \sum_{\substack{i \in V \\ i \neq j}} p_{ji}^{out} = 0$

Pipe capacity: $\forall i \neq j \in V, \quad p_{ij}^{in} \leq C_{ij}^{max} X_{ij}$

Source constraint: $\sum_{\substack{i \in V \\ i \neq v_0}} X_{i0} = 0$

Maximum flow constraint: $\sum_{\substack{j \in V \\ j \neq v_0}} p_{v_0 j}^{in} \leq Q^{max}$

Equity constraints:

- **Minimum distribution:** $\forall t, \quad \sum_{\substack{i,j \in V \\ i \neq j}} N_{jt} X_{ij} \geq \mu_t \sum_{\substack{i,j \in V \\ i \neq j}} X_{ij}$

- **Minimum surface threshold:**

We introduce: $\gamma S_{tot} = \gamma \sum_{i \in V} S_i \quad 0 < \gamma < 1$

The constraint becomes: $\sum_{j \in V} S_j X_{ij} \geq \gamma S_{tot}$

- **Minimum demand threshold:**

We introduce: $\gamma D_{tot} = \gamma \sum_{i \in V} D_i \quad 0 < \gamma < 1$

The constraint becomes: $\sum_{j \in V} D_j X_{ij} \geq \gamma D_{tot}$

Domains: $\begin{cases} X_{ij} \in \{0, 1\} & \forall i, j \in V \\ p_{ij}^{in}, p_{ij}^{out} \in \mathbb{R}^+ & \forall i, j \in V \end{cases}$